EXISTENCE OF A COMPLETE DEVISSAGE AFTER ONE ELEMENTARY ETALE LOCALIZATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; the Xy-pic xy environment is required because the counted proof displays two commutative diagrams. The native class line, the xr-hyper externaldocument block and the reference/slogan/history display environments are omitted because the assembler supplies a plain amsart wrapper and this excerpt uses none of them; no macro was invented or redefined.

Editorial note. The notion of a complete devissage is defined earlier in the same chapter and is declared as an allowed standard prerequisite instead of being dragged in as a context span, because the counted obligation is the existence and gluing argument, not the definition.

Allowed standard prerequisites (editorial).

- Complete devissage of a finite type module (paper's own notion). Same chapter, Definition environment labelled definition-complete-devissage, source line 1195; the pinned file defines it and it is quoted in the statement as the property being produced. Supplies the meaning of 'there exists a complete devissage of F_i/Y_i/S' over s" and the recognised data (Z_k, Y_k, i_k, pi_k, G_k, alpha_k); no part of the counted construction is taken from it.
- Elementary etale neighbourhood of a pointed scheme. Earlier chapters of the same project; imported by this chapter as standard input (Stacks tags 057K, 057L, 057R are the concrete predecessors used). Supplies the notion of the pointed morphism (S', s') -> (S, s) and the fact that its underlying morphism g : X' -> X is open, used at proof lines 1596-1601 to cover the fibre.

Source reference. Original lemma 05HJ: sources/stacks-flat/pinned-original-source.tex, source lines 1271–1271. Same-chapter navigation pointer to the base-change lemma invoked at proof line 1614 for the finitely many Y_i; the counted content of the proof spans (pointwise data, finite cover of X_s, fibre product of the bases) is retained verbatim in the delivered body, so this pointer replaces nothing.

Lemma 0.1. *Let $X \to S$ be a finite type morphism of schemes. Let $\mathcal{F}$ be a finite type quasi-coherent $\mathcal{O}_X$-module. Let $s \in S$ be a point. There exists an elementary étale neighbourhood $(S', s') \to (S, s)$ and étale morphisms $h_i : Y_i \to X_{S'}$, $i = 1, \ldots, n$ such that for each i there exists a complete dévissage of $\mathcal{F}_i/Y_i/S'$ over s', where $\mathcal{F}_i$ is the pullback of $\mathcal{F}$ to Y_i and such that $X_s = (X_{S'})_{s'} \subset \bigcup h_i(Y_i)$.*

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Proof. For every point $x \in X_s$ we can find a diagram

$$\begin{array}{ccc} (X, x) & \xleftarrow{g} & (X', x') \\ \downarrow & & \downarrow \\ (S, s) & \longleftarrow & (S', s') \end{array}$$

of pointed schemes such that the horizontal arrows are elementary étale neighbourhoods and such that $g^*\mathcal{F}/X'/S'$ has a complete dévissage at x'. As $X \to S$ is of finite type the fibre X_s is quasi-compact, and since each $g : X' \to X$ as above is open we can cover X_s by a finite union of $g(X'_{s'})$. Thus we can find a finite family of such diagrams

$$\begin{array}{ccc} (X, x) & \xleftarrow{g_i} & (X'_i, x'_i) \\ \downarrow & & \downarrow \\ (S, s) & \longleftarrow & (S'_i, s'_i) \end{array} \qquad i = 1, \ldots, n$$

such that $X_s = \bigcup g_i(X'_i)$. Set $S' = S'_1 \times_S \ldots \times_S S'_n$ and let $Y_i = X_i \times_{S'_i} S'$ be the base change of X'_i to S'. By Lemma 05HJ we see that the pullback of $\mathcal{F}$ to Y_i has a complete dévissage over s and we win. $\square$